NTA JEE Mains Jan 2026

Test Date	22/01/2026
Test Time	3:00 PM - 6:00 PM
Subject	B. Tech

Section : **Mathematics Section A**

Q.1 The area of the region $A = \{(x, y) : 4x^2 + y^2 \leq 8 \text{ and } y^2 \leq 4x\}$ is :

Options

1. $\pi + 4$
2. $\frac{\pi}{2} + \frac{1}{3}$
3. $\pi + \frac{2}{3}$
4. $\frac{\pi}{2} + 2$

Question Type : **MCQ**
Question ID : **860654994**
Option 1 ID : **8606543391**
Option 2 ID : **8606543388**
Option 3 ID : **8606543389**
Option 4 ID : **8606543390**

Q.2 Let S and S′ be the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ and $P(\alpha, \beta)$ be a point on the ellipse in the first quadrant. If $(SP)^2 + (S'P)^2 - SP \cdot S'P = 37$, then $\alpha^2 + \beta^2$ is equal to :

Options

1. 13
2. 15
3. 11
4. 17

Question Type : **MCQ**
Question ID : **860654985**
Option 1 ID : **8606543353**
Option 2 ID : **8606543354**
Option 3 ID : **8606543352**
Option 4 ID : **8606543355**

Q.3

If $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is a solution of the system of equations $AX = B$, where $\operatorname{adj} A = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$ and

$B = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$, then $|x+y+z|$ is equal to :

Options

1. 1
2. 3
3. 2
4. $\dfrac{3}{2}$

Question Type : **MCQ**
Question ID : **860654981**
Option 1 ID : **8606543336**
Option 2 ID : **8606543339**
Option 3 ID : **8606543338**
Option 4 ID : **8606543337**

Q.4

Let L be the line $\frac{x+1}{2}=\frac{y+1}{3}=\frac{z+3}{6}$ and let S be the set of all points (a, b, c) on L, whose distance from the line $\frac{x+1}{2}=\frac{y+1}{3}=\frac{z-9}{0}$ along the line L is 7. Then $\sum_{(a,\,b,\,c)\in S}(a+b+c)$ is equal to :

Options

1. 40
2. 34
3. 6
4. 28

Question Type : **MCQ**
Question ID : **860654990**
Option 1 ID : **8606543375**
Option 2 ID : **8606543374**
Option 3 ID : **8606543372**
Option 4 ID : **8606543373**

Q.5 Let the domain of the function $f(x) = \log_3\log_5\left(7 - \log_2\left(x^2 - 10x + 85\right)\right) + \sin^{-1}\left(\left|\frac{3x - 7}{17 - x}\right|\right)$ be $(\alpha, \beta]$. Then $\alpha + \beta$ is equal to :

Options

1. 12
2. 8
3. 10
4. 9

Question Type : **MCQ**
Question ID : **860654979**
Option 1 ID : **8606543331**
Option 2 ID : **8606543328**
Option 3 ID : **8606543330**
Option 4 ID : **8606543329**

Q.6 Let $f(x) = [x]^2 - [x+3] - 3$, $x \in \mathbf{R}$, where $[\cdot]$ is the greatest integer funtion. Then

Options

1. $f(x) = 0$ for finitely many values of x
2. $f(x) < 0$ only for $x \in [-1, 3)$
3. $f(x) > 0$ only for $x \in [4, \infty)$
4. $\int_0^2 f(x)\mathrm{d}x = -6$

Question Type : **MCQ**
Question ID : **860654993**
Option 1 ID : **8606543384**
Option 2 ID : **8606543385**
Option 3 ID : **8606543386**
Option 4 ID : **8606543387**

Q.7 Let f and g be functions satisfying $f(x+y) = f(x)f(y)$, $f(1) = 7$ and $g(x+y) = g(xy)$, $g(1) = 1$, for all $x, y \in \mathbf{N}$. If $\sum_{x=1}^{n}\left(\frac{f(x)}{g(x)}\right) = 19607$, then n is equal to :

Options

1. 5
2. 7
3. 4
4. 6

Question Type : **MCQ**
Question ID : **860654977**
Option 1 ID : **8606543321**
Option 2 ID : **8606543323**
Option 3 ID : **8606543320**
Option 4 ID : **8606543322**

Q.8 Let α, β be the roots of the quadratic equation $12x^2 - 20x + 3\lambda = 0$, $\lambda \in \mathbf{Z}$. If $\frac{1}{2} \leq |\beta - \alpha| \leq \frac{3}{2}$, then the sum of all possible values of λ is :

Options
1. 1
2. 3
3. 6
4. 4

Question Type : **MCQ**
Question ID : **860654978**
Option 1 ID : **8606543324**
Option 2 ID : **8606543325**
Option 3 ID : **8606543327**
Option 4 ID : **8606543326**

Q.9 If $y = y(x)$ satisfies the differential equation

$16\left(\sqrt{x + 9\sqrt{x}}\right)\left(4 + \sqrt{9 + \sqrt{x}}\right) \cos y \, dy = (1 + 2 \sin y)dx, x > 0$ and $y(256) = \frac{\pi}{2}$, $y(49) = \alpha$, then $2 \sin\alpha$ is equal to :

Options
1. $2\sqrt{2} - 1$
2. $2(\sqrt{2} - 1)$
3. $3(\sqrt{2} - 1)$
4. $\sqrt{2} - 1$

Question Type : **MCQ**
Question ID : **860654995**
Option 1 ID : **8606543392**
Option 2 ID : **8606543394**
Option 3 ID : **8606543395**
Option 4 ID : **8606543393**

Q.10 Let $P(10, 2\sqrt{15})$ be a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, whose foci are S and S′. If the length of its latus rectum is 8, then the square of the area of Δ PSS′ is equal to :

Options
1. 4200
2. 1462
3. 900
4. 2700

Question Type : **MCQ**
Question ID : **860654986**
Option 1 ID : **8606543359**
Option 2 ID : **8606543358**
Option 3 ID : **8606543356**
Option 4 ID : **8606543357**

Q.11 If $\lim_{x \to 0} \dfrac{e^{(a-1)x} + 2\cos bx + (c-2)e^{-x}}{x\cos x - \log_e(1+x)} = 2$, then $a^2+b^2+c^2$ is equal to :

Options
1. 9
2. 3
3. 7
4. 5

Question Type : **MCQ**
Question ID : **860654992**
Option 1 ID : **8606543383**
Option 2 ID : **8606543380**
Option 3 ID : **8606543382**
Option 4 ID : **8606543381**

Q.12 The number of elements in the relation $R = \{(x, y) : 4x^2 + y^2 < 52, x, y \in \mathbf{Z}\}$ is

Options
1. 89
2. 86
3. 67
4. 77

Question Type : **MCQ**
Question ID : **860654976**
Option 1 ID : **8606543319**
Option 2 ID : **8606543318**
Option 3 ID : **8606543316**
Option 4 ID : **8606543317**

Q.13 If the mean deviation about the median of the numbers k, 2k, 3k, ..., 1000k is 500, then k^2 is equal to :

Options
1. 16
2. 1
3. 4
4. 9

Question Type : **MCQ**
Question ID : **860654983**
Option 1 ID : **8606543345**
Option 2 ID : **8606543347**
Option 3 ID : **8606543346**
Option 4 ID : **8606543344**

Q.14 Let $[\cdot]$ denote the greatest integer function, and let $f(x) = \min\{\sqrt{2}x, x^2\}$.

Let $S = \{x \in (-2, 2) :$ the function $g(x) = |x|[x^2]$ is discontinuous at $x\}$.

Then $\sum_{x \in S} f(x)$ equals

Options

1. $2\sqrt{6} - 3\sqrt{2}$
2. $2 - \sqrt{2}$
3. $\sqrt{6} - 2\sqrt{2}$
4. $1 - \sqrt{2}$

Question Type : **MCQ**
Question ID : **860654991**
Option 1 ID : **8606543379**
Option 2 ID : **8606543376**
Option 3 ID : **8606543378**
Option 4 ID : **8606543377**

Q.15 Let $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \lambda\hat{j} + 2\hat{k}$, $\lambda \in \mathbf{Z}$ be two vectors. Let $\vec{c} = \vec{a} \times \vec{b}$ and $\vec{d}$ be a vector of magnitude 2 in yz-plane. If $|\vec{c}| = \sqrt{53}$, then the maximum possible value of $\left(\vec{c} \cdot \vec{d}\right)^2$ is equal to :

Options

1. 208
2. 52
3. 104
4. 26

Question Type : **MCQ**
Question ID : **860654989**
Option 1 ID : **8606543371**
Option 2 ID : **8606543369**
Option 3 ID : **8606543370**
Option 4 ID : **8606543368**

Q.16 Let C_r denote the coefficient of x^r in the binomial expansion of $(1+x)^n$, $n \in \mathbf{N}$, $0 \le r \le n$. If $P_n = C_0 - C_1 + \frac{2^2}{3}C_2 - \frac{2^3}{4}C_3 + \ldots + \frac{(-2)^n}{n+1}C_n$, then the value of $\sum_{n=1}^{25} \frac{1}{P_{2n}}$ equals.

Options
1. 525
2. 675
3. 650
4. 580

Question Type : **MCQ**
Question ID : **860654984**
Option 1 ID : **8606543349**
Option 2 ID : **8606543348**
Option 3 ID : **8606543350**
Option 4 ID : **8606543351**

Q.17 Among the statements

(S1) : If $A(5, -1)$ and $B(-2, 3)$ are two vertices of a triangle, whose orthocentre is $(0, 0)$, then its third vertex is $(-4, -7)$

and

(S2) : If positive numbers 2a, b, c are three consecutive terms of an A.P., then the lines $ax + by + c = 0$ are concurrent at $(2, -2)$,

Options
1. both are incorrect
2. only (S2) is correct
3. only (S1) is correct
4. both are correct

Question Type : **MCQ**
Question ID : **860654988**
Option 1 ID : **8606543365**
Option 2 ID : **8606543367**
Option 3 ID : **8606543366**
Option 4 ID : **8606543364**

Q.18 Let n be the number obtained on rolling a fair die. If the probability that the system

$$x - ny + z = 6$$
$$x + (n-2)y + (n+1)z = 8$$
$$(n-1)y + z = 1$$

has a unique solution is $\frac{k}{6}$, then the sum of k and all possible values of n is :

Options

1. 21
2. 24
3. 20
4. 22

Question Type : **MCQ**
Question ID : **860654982**
Option 1 ID : **8606543341**
Option 2 ID : **8606543343**
Option 3 ID : **8606543340**
Option 4 ID : **8606543342**

Q.19 Let the locus of the mid-point of the chord through the origin O of the parabola $y^2 = 4x$ be the curve S. Let P be any point on S. Then the locus of the point, which internally divides OP in the ratio 3:1, is :

Options

1. $3y^2 = 2x$
2. $3x^2 = 2y$
3. $2x^2 = 3y$
4. $2y^2 = 3x$

Question Type : **MCQ**
Question ID : **860654987**
Option 1 ID : **8606543361**
Option 2 ID : **8606543363**
Option 3 ID : **8606543362**
Option 4 ID : **8606543360**

Q.20

Let $S = \{z \in \mathbb{C} : 4z^2 + \bar{z} = 0\}$. Then $\sum_{z \in S} |z|^2$ is equal to :

Options

1. $\frac{1}{16}$
2. $\frac{7}{64}$
3. $\frac{5}{64}$
4. $\frac{3}{16}$

Question Type : **MCQ**
Question ID : **860654980**
Option 1 ID : **8606543334**
Option 2 ID : **8606543335**
Option 3 ID : **8606543332**
Option 4 ID : **8606543333**

Section : **Mathematics Section B**

Q.21

Let $\cos(\alpha + \beta) = -\frac{1}{10}$ and $\sin(\alpha - \beta) = \frac{3}{8}$, where $0 < \alpha < \frac{\pi}{3}$ and $0 < \beta < \frac{\pi}{4}$.

If $\tan 2\alpha = \frac{3(1 - r\sqrt{5})}{\sqrt{11}(s + \sqrt{5})}$, $r, s \in \mathbf{N}$, then $r+s$ is equal to ________.

Question Type : **SA**
Question ID : **860654998**

Q.22

Let S be the set of the first 11 natural numbers. Then the number of elements in $A = \{B \subseteq S : n(B) \geq 2$ and the product of all elements of B is even$\}$ is ________.

Question Type : **SA**
Question ID : **860654997**

Q.23

Let $[\cdot]$ be the greatest integer function. If $\alpha = \int_0^{64} \left(x^{1/3} - [x^{1/3}]\right) dx$, then $\frac{1}{\pi} \int_0^{\alpha\pi} \left(\frac{\sin^2\theta}{\sin^6\theta + \cos^6\theta} \right) d\theta$ is equal to ________.

Question Type : **SA**
Question ID : **8606541000**

Q.24

Suppose a, b, c are in A.P. and $a^2, 2b^2, c^2$ are in G.P. If $a < b < c$ and $a + b + c = 1$, then $9(a^2 + b^2 + c^2)$ is equal to ________.

Question Type : **SA**
Question ID : **860654996**

Q.25 Let a vector $\vec{a} = \sqrt{2}\hat{i} - \hat{j} + \lambda\hat{k}$, $\lambda > 0$, make an obtuse angle with the vector $\vec{b} = -\lambda^2\hat{i} + 4\sqrt{2}\hat{j} + 4\sqrt{2}\hat{k}$ and an angle θ, $\frac{\pi}{6} < \theta < \frac{\pi}{2}$, with the positive z-axis. If the set of all possible values of λ is $(\alpha, \beta) - \{\gamma\}$, then $\alpha + \beta + \gamma$ is equal to ________.

Question Type : **SA**
Question ID : **860654999**

Section : **Physics Section A**

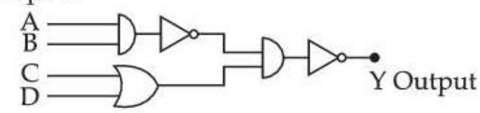
Q.26 Light is incident on a metallic plate having work function 110×10^{-20} J. If the produced photoelectrons have zero kinetic energy then the angular frequency of the incident light is ________rad/s. $(h = 6.63 \times 10^{-34}$ J.s).

Options

1. 1.66×10^{15}
2. 1.04×10^{13}
3. 1.04×10^{16}
4. 1.66×10^{16}

Question Type : **MCQ**
Question ID : **8606541018**
Option 1 ID : **8606543470**
Option 2 ID : **8606543469**
Option 3 ID : **8606543472**
Option 4 ID : **8606543471**

Q.27 The correct truth table for the given input data of the following logic gate is :



Options

1.

Inputs				Output
A	B	C	D	Y
1	1	0	1	1
0	0	1	1	0
1	0	1	0	1
1	1	1	1	0

2.

Inputs				Output
A	B	C	D	Y
1	1	0	1	0
0	0	1	1	1
1	0	1	0	1
1	1	1	1	1

3.

Inputs				Output
A	B	C	D	Y
1	1	0	1	1
0	0	1	1	0
1	0	1	0	0
1	1	1	1	1

4.

Inputs				Output
A	B	C	D	Y
1	1	0	1	0
0	0	1	1	0
1	0	1	0	1
1	1	1	1	1

Question Type : **MCQ**
Question ID : **8606541020**
Option 1 ID : **8606543477**
Option 2 ID : **8606543480**
Option 3 ID : **8606543479**
Option 4 ID : **8606543478**

Q.28 In an open organ pipe ν_3 and ν_6 are 3^{rd} and 6^{th} harmonic frequencies, respectively. If $\nu_6 - \nu_3 = 2200$ Hz then length of the pipe is __________ mm .
(Take velocity of sound in air is 330 m/s.)

Options
1. 250
2. 225
3. 200
4. 275

Question Type : **MCQ**
Question ID : **8606541011**
Option 1 ID : **8606543443**
Option 2 ID : **8606543442**
Option 3 ID : **8606543441**
Option 4 ID : **8606543444**

Q.29 The wavelength of light, while it is passing through water is 540 nm. The refractive index of water is $4/3$. The wavelength of the same light when it is passing through a transparent medium having refractive index of $3/2$ is _________ nm.

Options
1. 840
2. 540
3. 380
4. 480

Question Type : **MCQ**
Question ID : **8606541015**
Option 1 ID : **8606543458**
Option 2 ID : **8606543459**
Option 3 ID : **8606543460**
Option 4 ID : **8606543457**

Q.30 When a part of a straight capillary tube is placed vertically in a liquid, the liquid raises upto certain height h. If the inner radius of the capillary tube, density of the liquid and surface tension of the liquid decrease by 1% each, then the height of the liquid in the tube will change by ______ %.

Options
1. -1
2. -3
3. $+1$
4. $+3$

Question Type : **MCQ**
Question ID : **8606541007**
Option 1 ID : **8606543426**
Option 2 ID : **8606543428**
Option 3 ID : **8606543425**
Option 4 ID : **8606543427**

Q.31 Given below are two statements :

Statement I : For a mechanical system of many particles total kinetic energy is the sum of kinetic energies of all the particles.

Statement II : The total kinetic energy can be the sum of kinetic energy of the center of mass w.r.t to the origin and the kinetic energy of all the particles w.r.t. the center of mass as the reference.

In the light of the above statements, choose the **correct answer** from the options given below :

Options

1. Both **Statement I** and **Statement II** are false
2. **Statement I** is false but **Statement II** is true
3. Both **Statement I** and **Statement II** are true
4. **Statement I** is true but **Statement II** is false

Question Type : **MCQ**
Question ID : **8606541005**
Option 1 ID : **8606543418**
Option 2 ID : **8606543420**
Option 3 ID : **8606543417**
Option 4 ID : **8606543419**

Q.32 The smallest wavelength of Lyman series is 91 nm. The difference between the largest wavelengths of *Paschen* and *Balmer* series is nearly ______ nm.

Options

1. 1784
2. 1550
3. 1875
4. 1217

Question Type : **MCQ**
Question ID : **8606541019**
Option 1 ID : **8606543475**
Option 2 ID : **8606543473**
Option 3 ID : **8606543474**
Option 4 ID : **8606543476**

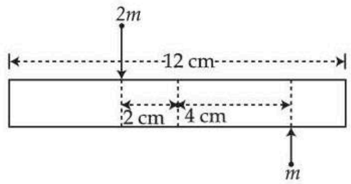
Q.33 If ϵ, E and t represent the free space permittivity, electric field and time respectively, then the unit of $\frac{\epsilon E}{t}$ will be :

Options

1. Am^2
2. A/m
3. A/m^2
4. Am

Question Type : **MCQ**
Question ID : **8606541002**
Option 1 ID : **8606543406**
Option 2 ID : **8606543408**
Option 3 ID : **8606543405**
Option 4 ID : **8606543407**

Q.34 A uniform bar of length 12 cm and mass 20 m lies on a smooth horizontal table. Two point masses m and $2m$ are moving in opposite directions with same speed of v and in the same plane as the bar, as shown in figure. These masses strike the bar simultaneously and get stuck to it. After collision the entire system is rotating with angular frequency ω. The ratio of v and ω is :

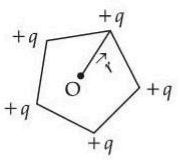


Options

1. 66
2. 32
3. 33
4. $2\sqrt{88}$

Question Type : **MCQ**
Question ID : **8606541003**
Option 1 ID : **8606543412**
Option 2 ID : **8606543409**
Option 3 ID : **8606543410**
Option 4 ID : **8606543411**

Q.35 Five positive charges each having charge q are placed at the vertices of a pentagon as shown in the figure. The electric potential (V) and the electric field $(\vec{E})$ at the center O of the pentagon due to these five positive charges are :



Options

1. $V=\dfrac{5q}{4\pi\varepsilon_0 r}$ and $\vec{E}=\dfrac{5q}{4\pi\varepsilon_0 r^2}\hat{r}$
2. $V=\dfrac{5q}{4\pi\varepsilon_0 r}$ and $\vec{E}=\dfrac{5\sqrt{3}q}{8\pi\varepsilon_0 r^2}\hat{r}$
3. $V=0$ and $\vec{E}=0$
4. $V=\dfrac{5q}{4\pi\varepsilon_0 r}$ and $\vec{E}=0$

Question Type : **MCQ**
Question ID : **8606541012**
Option 1 ID : **8606543446**
Option 2 ID : **8606543448**
Option 3 ID : **8606543445**
Option 4 ID : **8606543447**

Q.36 A laser beam has intensity of $4.0 \times 10^{14}\ \mathrm{W/m^2}$. The amplitude of magnetic field associated with beam is __________T. (Take $\epsilon_o = 8.85 \times 10^{-12}\ \mathrm{C^2/Nm^2}$ and $c = 3 \times 10^8\ \mathrm{m/s}$)

Options

1. 18.3
2. 5.5
3. 1.83
4. 2.0

Question Type : **MCQ**
Question ID : **8606541014**
Option 1 ID : **8606543454**
Option 2 ID : **8606543453**
Option 3 ID : **8606543455**
Option 4 ID : **8606543456**

Q.37 Given below are two statements :

Statement I : A satellite is moving around earth in the orbit very close to the earth surface. The time period of revolution of satellite depends upon the density of earth.

Statement II : The time period of revolution of the satellite is $T = 2\pi\sqrt{\frac{R_e}{g}}$ (for satellite very close to the earth surface), where R_e radius of earth and g acceleration due to gravity.

In the light of the above statements, choose the **correct** answer from the options given below :

Options

1. **Statement I** is false but **Statement II** is true
2. Both **Statement I** and **Statement II** are true
3. **Statement I** is true but **Statement II** is false
4. Both **Statement I** and **Statement II** are false

Question Type : **MCQ**
Question ID : **8606541006**
Option 1 ID : **8606543424**
Option 2 ID : **8606543421**
Option 3 ID : **8606543423**
Option 4 ID : **8606543422**

Q.38 In parallax method for the determination of focal length of a concave mirror, the object should always be placed :

Options

1. between the pole(P) and the focus(F) of the concave mirror ONLY

2. beyond the centre of the curvature(C) of the mirror ONLY

3. at any point beyond the focus(F) of the mirror

4. between the focus(F) and the centre of curvature(C) of the mirror ONLY

Question Type : **MCQ**
Question ID : **8606541016**
Option 1 ID : **8606543461**
Option 2 ID : **8606543463**
Option 3 ID : **8606543464**
Option 4 ID : **8606543462**

Q.39 Three small identical bubbles of water having same charge on each coalesce to form a bigger bubble. Then the ratio of the potentials on one initial bubble and that on the resultant bigger bubble is :

Options

1. $1 : 2^{2/3}$

2. $1 : 3^{2/3}$

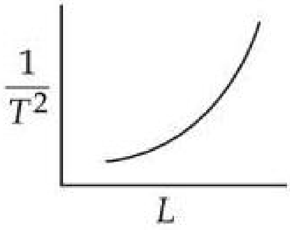
3. $1 : 3^{1/3}$

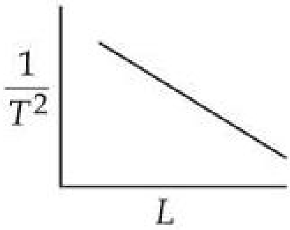
4. $3^{2/3} : 1$

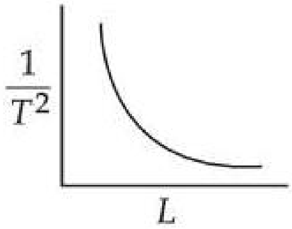
Question Type : **MCQ**
Question ID : **8606541008**
Option 1 ID : **8606543431**
Option 2 ID : **8606543429**
Option 3 ID : **8606543430**
Option 4 ID : **8606543432**

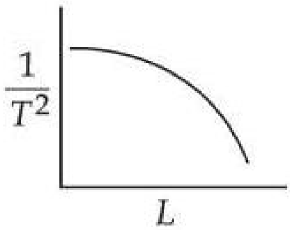
Q.40 Using a simple pendulum experiment g is determind by measuring its time period T. Which of the following plots represent the correct relation between the pendulum length L and time period T?

Options

1. 

2. 

3. 

4. 

Question Type : **MCQ**
Question ID : **8606541001**
Option 1 ID : **8606543402**
Option 2 ID : **8606543404**
Option 3 ID : **8606543401**
Option 4 ID : **8606543403**

Q.41 Consider two boxes containing ideal gases A and B such that their temperatures, pressures and number densities are same. The molecular size of A is half of that of B and mass of molecule A is four times that of B. If the collision frequency in gas B is 32×10^{18}/s then collision frequency in gas A is __________/s.

Options

1. 32×10^{8}
2. 4×10^{8}
3. 8×10^{8}
4. 2×10^{8}

Question Type : **MCQ**
Question ID : **8606541010**
Option 1 ID : **8606543440**
Option 2 ID : **8606543438**
Option 3 ID : **8606543439**
Option 4 ID : **8606543437**

Q.42 Which of the following are true for a single slit diffraction ?

A. Width of central maxima increases with increase in wavelength keeping slit width constant.

B. Width of central maxima increases with decrease in wavelength keeping slit width constant.

C. Width of central maxima increases with decrease in slit width at constant wavelength.

D. Width of central maxima increases with increase in slit width at constant wavelength.

E. Brightness of central maxima increases for decrease in wavelength at constant slit width.

Options

1. B, D only
2. A, D only
3. A, D, E only
4. B, C only

Question Type : **MCQ**
Question ID : **8606541017**
Option 1 ID : **8606543467**
Option 2 ID : **8606543466**
Option 3 ID : **8606543465**
Option 4 ID : **8606543468**

Q.43 Given below are two statements :

Statement I : An object moves from position r_1 to position r_2 under a conservative force field $\vec{F}$.

The work done by the force is $W = -\int_{r_1}^{r_2} \vec{F} \cdot \vec{dr}$.

Statement II : Any object moving from one location to another location can follow infinite number of paths. Therefore, the amount of work done by the object changes with the path it follows for a conservative force.

In the light of the above statements, choose the **correct answer** from the options given below :

Options

1. Both **Statement I** and **Statement II** are true
2. Both **Statement I** and **Statement II** are false
3. **Statement I** is true but **Statement II** is false
4. **Statement I** is false but **Statement II** is true

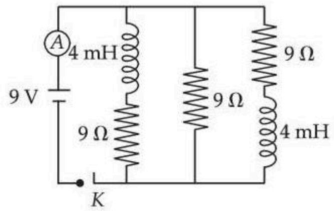
Question Type : **MCQ**
Question ID : **8606541004**
Option 1 ID : **8606543413**
Option 2 ID : **8606543414**
Option 3 ID : **8606543415**
Option 4 ID : **8606543416**

Q.44 An electric power line having total resistance of 2 Ω, delivers 1 kW of power at 250 V. The percentage efficiency of transmission line is ________.

Options
1. 86.5
2. 100
3. 92.5
4. 96.9

Question Type : **MCQ**
Question ID : **8606541013**
Option 1 ID : **8606543450**
Option 2 ID : **8606543449**
Option 3 ID : **8606543451**
Option 4 ID : **8606543452**

Q.45 Figure shows the circuit that contains three resistances (9 Ω each) and two inductors (4 mH each). The reading of ammeter at the moment switch K is turned ON, is ________A.



Options
1. 3
2. zero
3. 2
4. 1

Question Type : **MCQ**
Question ID : **8606541009**
Option 1 ID : **8606543435**
Option 2 ID : **8606543436**
Option 3 ID : **8606543434**
Option 4 ID : **8606543433**

Section : **Physics Section B**

Q.46 A capacitor P with capacitance 10×10^{-6} F is fully charged with a potential difference of 6.0 V and disconnected from the battery. The charged capacitor P is connected across another capacitor Q with capacitance 20×10^{-6} F. The charge on capacitor Q when equilibrium is established will be $\alpha \times 10^{-5}$ C (assume capacitor Q does not have any charge initially), the value of α is ________.

Question Type : **SA**
Question ID : **8606541022**

Q.47 An insulated cylinder of volume 60 cm^3 is filled with a gas at 27 °C and 2 atmospheric pressure. Then the gas is compressed making the final volume as 20 cm^3 while allowing the temperature to rise to 77 °C. The final pressure is ________ atmospheric pressure.

Question Type : **SA**
Question ID : **8606541024**

Q.48 A cylindrical conductor of length 2 m and area of cross-section 0.2 mm^2 carries an electric current of 1.6 A when its ends are connected to a 2 V battery. Mobility of electrons in the conductor is $\alpha \times 10^{-3}$ $m^2/V.s$. The value of α is :
(electron concentration $= 5 \times 10^{28}/m^3$ and electron charge $= 1.6 \times 10^{-19}$ C)

Question Type : **SA**
Question ID : **8606541023**

Q.49 Two masses m and 2m are connected by a light string going over a pulley (disc) of mass 30m with radius $r = 0.1$ m. The pulley is mounted in a vertical plane and it is free to rotate about its axis. The 2m mass is released from rest and its speed when it has descended through a height of 3.6 m is ________ m/s. (Assume string does not slip and $g = 10$ m/s^2)

Question Type : **SA**
Question ID : **8606541021**

Q.50 A conducting circular loop is rotated about its diameter at a constant angular speed of 100 rad/s in a magnetic field of 0.5 T perpendicular to the axis of rotation. When the loop is rotated by 30° from the horizontal position, the induced EMF is 15.4 mV. The radius of the loop is ________ mm.
$\left(\text{Take } \pi = \frac{22}{7}\right)$

Question Type : **SA**
Question ID : **8606541025**

Section : **Chemistry Section A**

Q.51 Among H_2S, H_2O, NF_3, NH_3 and $CHCl_3$, identify the molecule (X) with lowest dipole moment value. The number of lone pairs of electrons present on the central atom of the molecule (X) is :

Options
1. 3
2. 2
3. 1
4. 0

Question Type : **MCQ**
Question ID : **8606541032**
Option 1 ID : **8606543513**
Option 2 ID : **8606543512**
Option 3 ID : **8606543511**
Option 4 ID : **8606543510**

Q.52 Given below are two statements :

Statement I : $C < O < N < F$ is the correct order in terms of first ionization enthalpy values.

Statement II : $S > Se > Te > Po > O$ is the correct order in terms of the magnitude of electron gain enthalpy values.

In the light of the above statements, choose the **correct** answer from the options given below :

Options

1. Both **Statement I** and **Statement II** are true
2. **Statement I** is false but **Statement II** is true
3. **Statement I** is true but **Statement II** is false
4. Both **Statement I** and **Statement II** are false

Question Type : **MCQ**
Question ID : **8606541033**
Option 1 ID : **8606543514**
Option 2 ID : **8606543517**
Option 3 ID : **8606543516**
Option 4 ID : **8606543515**

Q.53 Given below are two statements :

Statement I : The first ionization enthalpy of Cr is lower than that of Mn.

Statement II : The second and third ionization enthalpies of Cr are higher than those of Mn.

In the light of the above statements, choose the **correct** answer from the options given below :

Options

1. Both **Statement I** and **Statement II** are false
2. **Statement I** is false but **Statement II** is true
3. Both **Statement I** and **Statement II** are true
4. **Statement I** is true but **Statement II** is false

Question Type : **MCQ**
Question ID : **8606541035**
Option 1 ID : **8606543523**
Option 2 ID : **8606543525**
Option 3 ID : **8606543522**
Option 4 ID : **8606543524**

Q.54 At T(K), 100 g of 98% H_2SO_4 (w/w) aqueous solution is mixed with 100 g of 49% H_2SO_4 (w/w) aqueous solution. What is the mole fraction of H_2SO_4 in the resultant solution ?
(Given : Atomic mass $H = 1$ u ; $S = 32$ u ; $O = 16$ u).
(Assume that temperature after mixing remains constant)

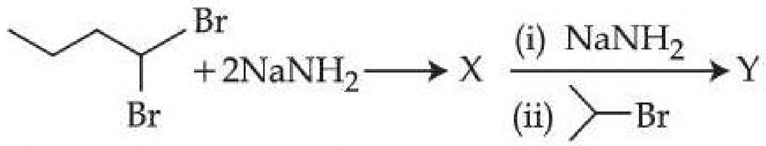
Options

1. 0.337
2. 0.1
3. 0.9
4. 0.663

Question Type : **MCQ**
Question ID : **8606541028**
Option 1 ID : **8606543497**
Option 2 ID : **8606543495**
Option 3 ID : **8606543496**
Option 4 ID : **8606543494**

Q.55

Consider the following reaction :



The product Y formed is :

Options

1. 2-methylhex-2-yne
2. Isopropylbut-1-yne
3. 2-methylhex-3-yne
4. 5-methylhex-2-yne

Question Type : **MCQ**

Question ID : **8606541040**

Option 1 ID : **8606543544**

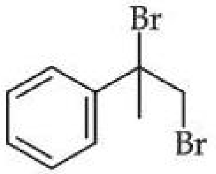
Option 2 ID : **8606543545**

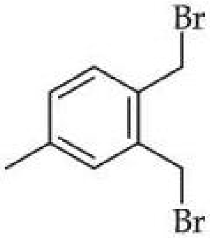
Option 3 ID : **8606543543**

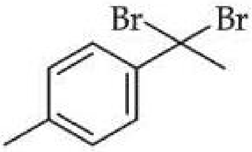
Option 4 ID : **8606543542**

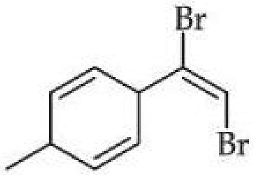
Q.56 The dibromo compound [P] (molecular formula : $C_9H_{10}Br_2$) when heated with excess sodamide followed by treatment with dilute HCl gives [Q]. On warming [Q] with mercuric sulphate and dilute sulphuric acid yield [R] which gives positive Iodoform test but negative Tollen's test. The compound [P] is :

Options

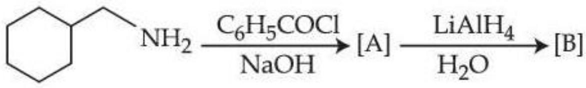
1. 

2. 

3. 

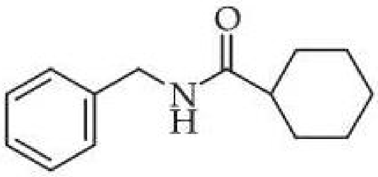
4. 

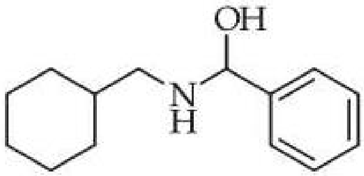
Question Type :	**MCQ**
Question ID :	**8606541041**
Option 1 ID :	**8606543546**
Option 2 ID :	**8606543548**
Option 3 ID :	**8606543547**
Option 4 ID :	**8606543549**

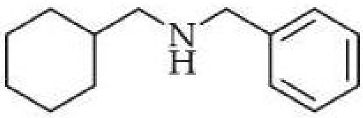
Q.57 

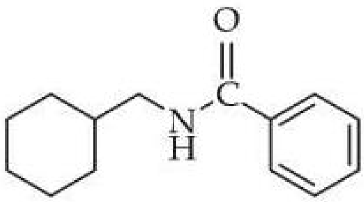
The final product [B] is :

Options

1. 

2. 

3. 

4. 

Question Type : **MCQ**
Question ID : **8606541044**
Option 1 ID : **8606543558**
Option 2 ID : **8606543560**
Option 3 ID : **8606543561**
Option 4 ID : **8606543559**

Q.58 $A + 2B \longrightarrow AB_2$

36.0 g of 'A' (Molar mass : 60 g mol^{-1}) and 56.0 g of 'B' (Molar mass : 80 g mol^{-1}) are allowed to react. Which of the following statements are **correct** ?

A. 'A' is the limiting reagent.

B. 77.0 g of AB_2 is formed.

C. Molar mass of AB_2 is 140 g mol^{-1}.

D. 15.0 g of A is left unreacted after the completion of reaction.

Choose the **correct** answer from the options given below :

Options

1. B and D Only
2. A and C Only
3. C and D Only
4. A and B Only

Question Type : **MCQ**
Question ID : **8606541026**
Option 1 ID : **8606543487**
Option 2 ID : **8606543489**
Option 3 ID : **8606543488**
Option 4 ID : **8606543486**

Q.59 When 1 g of compound (X) is subjected to Kjeldahl's method for estimation of nitrogen, 15 mL 1 M H_2SO_4 was neutralized by ammonia evolved. The percentage of nitrogen in compound (X) is :

Options
1. 42
2. 0.21
3. 0.42
4. 21

Question Type : **MCQ**
Question ID : **8606541038**
Option 1 ID : **8606543536**
Option 2 ID : **8606543534**
Option 3 ID : **8606543537**
Option 4 ID : **8606543535**

Q.60 Match **List - I** with **List - II**.

	List - I **Reaction of Glucose with**		**List - II** **Product formed**
A.	Hydroxylamine	I.	Gluconic acid
B.	Br_2 water	II.	Glucose pentacetate
C.	Excess acetic anhydride	III.	Saccharic acid
D.	Concentrated HNO_3	IV.	Glucoxime

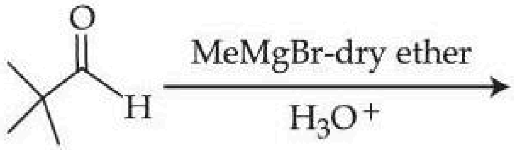
Choose the **correct** answer from the options given below :

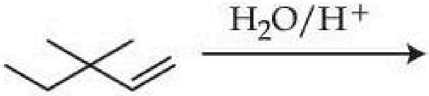
Options
1. A-IV, B-I, C-II, D-III
2. A-III, B-I, C-IV, D-II
3. A-IV, B-III, C-II, D-I
4. A-I, B-III, C-IV, D-II

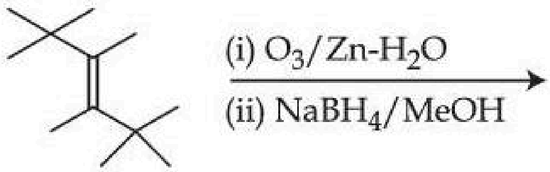
Question Type : **MCQ**
Question ID : **8606541045**
Option 1 ID : **8606543562**
Option 2 ID : **8606543564**
Option 3 ID : **8606543563**
Option 4 ID : **8606543565**

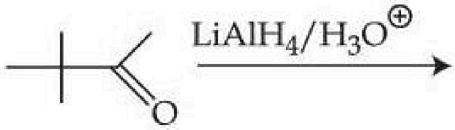
Q.61

3, 3-Dimethyl-2-butanol **cannot** be prepared by :

A. 

B. 

C. 

D. 

E. 

Choose the **correct** answer from the options given below :

Options

1. B, C and E Only
2. B and E Only
3. B Only
4. B and C Only

Question Type :	**MCQ**
Question ID :	**8606541043**
Option 1 ID :	**8606543556**
Option 2 ID :	**8606543557**
Option 3 ID :	**8606543555**
Option 4 ID :	**8606543554**

Q.62 Which of the following mixture gives a buffer solution with $pH = 9.25$?
Given : $pK_b\ (NH_4OH) = 4.75$

Options
1. $0.5\ M\ NH_4OH\ (0.2\ L) + 0.2\ M\ HCl\ (0.5\ L)$
2. $0.2\ M\ NH_4OH\ (0.5\ L) + 0.1\ M\ HCl\ (0.5\ L)$
3. $0.2\ M\ NH_4OH\ (0.4\ L) + 0.1\ M\ HCl\ (1\ L)$
4. $0.4\ M\ NH_4OH\ (1\ L) + 0.1\ M\ HCl\ (1\ L)$

Question Type : **MCQ**
Question ID : **8606541029**
Option 1 ID : **8606543501**
Option 2 ID : **8606543500**
Option 3 ID : **8606543499**
Option 4 ID : **8606543498**

Q.63 Given below are two statements :

Statement I : Elements 'X' and 'Y' are the most and least electronegative elements, respectively among N, As, Sb and P. The nature of the oxides X_2O_3 and Y_2O_3 is acidic and amphoteric, respectively.

Statement II : BCl_3 is covalent in nature and gets hydrolysed in water. It produces $[B(OH)_4]^-$ and $[B(H_2O)_6]^{3+}$ in aqueous medium.

In the light of the above statements, choose the **correct** answer from the options given below :

Options
1. **Statement I** is false but **Statement II** is true
2. Both **Statement I** and **Statement II** are true
3. Both **Statement I** and **Statement II** are false
4. **Statement I** is true but **Statement II** is false

Question Type : **MCQ**
Question ID : **8606541034**
Option 1 ID : **8606543521**
Option 2 ID : **8606543518**
Option 3 ID : **8606543519**
Option 4 ID : **8606543520**

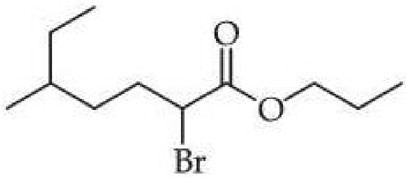
Q.64 The compound A, $C_8H_8O_2$ reacts with acetophenone to form a single product via cross-Aldol condensation. The compound A on reaction with conc. NaOH forms a substituted benzyl alcohol as one of the two products. The compound A is :

Options

1. 4-methoxy benzaldehyde
2. 4-hydroxy benzylaldehyde
3. 4-methyl benzoic acid
4. 2-hydroxy acetophenone

Question Type : **MCQ**
Question ID : **8606541042**
Option 1 ID : **8606543550**
Option 2 ID : **8606543551**
Option 3 ID : **8606543553**
Option 4 ID : **8606543552**

Q.65 The IUPAC name of the following compound is :



Options

1. n-propyl-2-bromo-5-methylheptanoate
2. 2-bromo-5-methylhexylpropanoate
3. n-propyl-1-bromo-4-methylhexanoate
4. 2-bromo-5-methylpropanoate

Question Type : **MCQ**
Question ID : **8606541039**
Option 1 ID : **8606543539**
Option 2 ID : **8606543541**
Option 3 ID : **8606543540**
Option 4 ID : **8606543538**

Q.66 Identify the **correct** statements :

A. Hydrated salts can be used as primary standard.

B. Primary standard should not undergo any reaction with air.

C. Reactions of primary standard with another substance should be instantaneous and stoichiometric.

D. Primary standard should not be soluble in water.

E. Primary standard should have low relative molar mass.

Choose the **correct** answer from the options given below :

Options

1. A, B, C and E Only
2. A, B and C Only
3. A, B and E Only
4. D and E Only

Question Type : **MCQ**
Question ID : **8606541037**
Option 1 ID : **8606543530**
Option 2 ID : **8606543532**
Option 3 ID : **8606543531**
Option 4 ID : **8606543533**

Q.67 Consider the following reduction processes :

$Al^{3+} + 3e^- \longrightarrow Al(s), E^o = -1.66\ V$

$Fe^{3+} + e^- \longrightarrow Fe^{2+}, E^o = +0.77\ V$

$Co^{3+} + e^- \longrightarrow Co^{2+}, E^o = +1.81\ V$

$Cr^{3+} + 3e^- \longrightarrow Cr(s), E^o = -0.74\ V$

The tendency to act as reducing agent decreases in the order :

Options

1. $Cr > Fe^{2+} > Al > Co^{2+}$
2. $Al > Cr > Fe^{2+} > Co^{2+}$
3. $Al > Fe^{2+} > Cr > Co^{2+}$
4. $Al > Cr > Co^{2+} > Fe^{2+}$

Question Type : **MCQ**
Question ID : **8606541030**
Option 1 ID : **8606543505**
Option 2 ID : **8606543503**
Option 3 ID : **8606543504**
Option 4 ID : **8606543502**

Q.68 The energy of first (lowest) Balmer line of H atom is x J. The energy (in J) of second Balmer line of H atom is :

Options

1. $2x$
2. x^2
3. $\frac{x}{1.35}$
4. $1.35x$

Question Type : **MCQ**
Question ID : **8606541027**
Option 1 ID : **8606543490**
Option 2 ID : **8606543493**
Option 3 ID : **8606543492**
Option 4 ID : **8606543491**

Q.69 $[Ni(PPh_3)_2Cl_2]$ is a paramagnetic complex. Identify the **INCORRECT** statements about this complex.

A. The complex exhibits geometrical isomerism.
B. The complex is white in colour.
C. The calculated spin-only magnetic moment of the complex is 2.84 BM.
D. The calculated CFSE (Crystal Field Stabilization Energy) of Ni in this complex is $-0.8\ \Delta_o$.
E. The geometrical arrangement of ligands in this complex is similar to that in $Ni(CO)_4$.

Choose the **correct** answer from the options given below :

Options

1. C, D and E Only
2. A, B and D Only
3. C and D Only
4. A and B Only

Question Type : **MCQ**
Question ID : **8606541036**
Option 1 ID : **8606543529**
Option 2 ID : **8606543526**
Option 3 ID : **8606543527**
Option 4 ID : **8606543528**

Q.70 Correct statements regarding Arrhenius equation among the following are :

A. Factor $e^{-Ea/RT}$ corresponds to fraction of molecules having kinetic energy less than Ea.

B. At a given temperature, lower the Ea, faster is the reaction.

C. Increase in temperature by about 10°C doubles the rate of reaction.

D. Plot of log k vs $\frac{1}{T}$ gives a straight line with slope $= -\frac{Ea}{R}$.

Choose the **correct** answer from the options given below :

Options

1. A and C Only
2. A and B Only
3. B and C Only
4. B and D Only

Question Type : **MCQ**
Question ID : **8606541031**
Option 1 ID : **8606543506**
Option 2 ID : **8606543509**
Option 3 ID : **8606543508**
Option 4 ID : **8606543507**

Section : **Chemistry Section B**

Q.71 Among the following oxides of 3d elements, the number of mixed oxides are ________.
Ti_2O_3, V_2O_4, Cr_2O_3, Mn_3O_4, Fe_3O_4, Fe_2O_3, Co_3O_4

Question Type : **SA**
Question ID : **8606541049**

Q.72 If the enthalpy of sublimation of Li is 155 kJ mol^{-1}, enthalpy of dissociation of F_2 is 150 kJ mol^{-1}, ionization enthalpy of Li is 520 kJ mol^{-1}, electron gain enthalpy of F is -313 kJ mol^{-1}, standard enthalpy of formation of LiF is -594 kJ mol^{-1}. The magnitude of lattice enthalpy of LiF is ________ kJ mol^{-1}. (Nearest Integer)

Question Type : **SA**
Question ID : **8606541046**

Q.73 Consider $A \xrightarrow{k_1} B$ and $C \xrightarrow{k_2} D$ are two reactions. If the rate constant (k_1) of the $A \longrightarrow B$ reaction can be expressed by the following equation $\log_{10} k = 14.34 - \frac{1.5 \times 10^4}{T/K}$ and activation energy of $C \longrightarrow D$ reaction (Ea_2) is $\frac{1}{5}$th of the $A \longrightarrow B$ reaction (Ea_1), then the value of (Ea_2) is ________ kJ mol^{-1}. (Nearest Integer)

Question Type : **SA**
Question ID : **8606541048**

Q.74 Consider the following electrochemical cell :

$$Pt \mid O_2(g)(1bar) \mid HCl(aq) \parallel M^{2+}(aq, 1.0\,M) \mid M(s)$$

The pH above which, oxygen gas would start to evolve at anode is ________ (nearest integer).

$$\left[\begin{array}{ll} \text{Given :} & \left.\begin{array}{l} E^o_{M^{2+}/M} = 0.994\,V \\ E^o_{O_2/H_2O} = 1.23\,V \end{array}\right\} \text{standard reduction potential} \\ & \text{and } \frac{RT}{F}(2.303) = 0.059\,V \text{ at the given condition} \end{array}\right]$$

Question Type : **SA**
Question ID : **8606541047**

Q.75 The mass of benzanilide obtained from the benzoylation reaction of 5.8 g of aniline, if yield of product is 82%, is ________ g (nearest integer).
(Given molar mass in g mol^{-1} H : 1, C : 12, N : 14, O : 16)

Question Type : **SA**
Question ID : **8606541050**

"